19 200 people were asked if they like any of running, cycling or swimming.

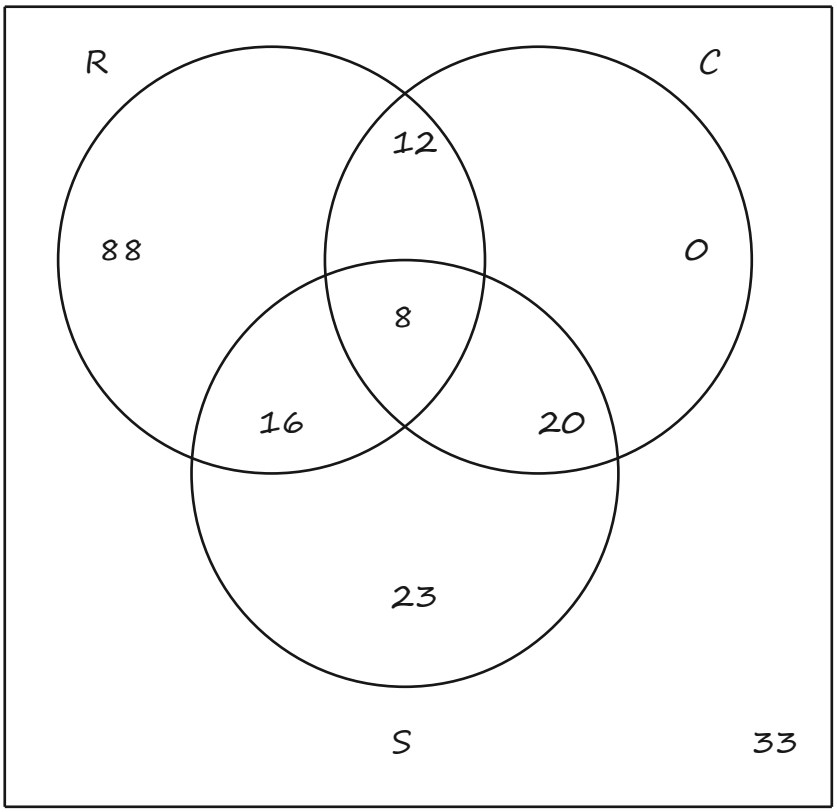
Of these people,

8 like running, swimming and cycling
16 like running and swimming but not cycling
33 do not like any of these activities
28 like cycling and swimming
all 40 people who like cycling like at least one other activity
124 like running

One of the people asked is chosen at random.

Given that this person likes swimming, find the probability that this person likes cycling.

C and S only = 28 – 8 = 20; C and R only = 40 – 28 = 12



1 mark
C and S only:
28 – 8 = 20.

1 mark
C only = 0.
R and C only = 12.
R only = 124 – 36 = 88.

R only = 124 – 12 – 16 – 8 = 88

S only = 200 – 33 – 88 – 12 – 16 – 8 – 20 = 23

1 mark
Subtract all other regions.

Total swimming = 16 + 8 + 20 + 23 = 67

1 mark
Use only swimmers.

$$P(\text{cycling} \mid \text{swimming}) = \frac{8 + 20}{67} = \frac{28}{67}$$

1 mark
28/67.

(Total for Question 19 is 5 marks)

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